

SLEEVEDRIFT · DESIGN DECISIONS REGISTER

SleeveDrift design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Round 2. Amish decided R3 (1B), R9 (2A) and the R1 stretch (3C) as recommended (SVD-DDR-003); two new questions raised while carrying them out, proposed, awaiting Amish
0.1	2026-10-03	Amish Chadha	Register opened; design decisions made under Amish's pre-approvals of 2026-10-03; three requirement decisions proposed for Amish

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` or in `docs/REVIEW.md`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below set SleeveDrift's safety case for people working inside a rescue pipe (overload limit, gas monitoring, air, ring fixing, casualty evacuation). Each took the conservative option; the evidence that would relax it is in SVD-DDR-001, Table 1.

■ Open decisions

The three round 1 requirement decisions (R3, R9, R1 stretch) were decided by Amish on 2026-10-03 and are listed under Decisions made (SVD-DDR-003). The questions below were raised while carrying them out (full statement in `docs/REVIEW.md`, session 2026-10-03, round 2). Each is **Proposed, awaiting Amish**.

#	To decide	Options (effect on the requirement, cost, mass)	Recommendation	What it affects in the build	Source
4	Drill gear and clutch rule (R11 with the drill drive): a clutch set between about 28 and 45 N m slips before the 2.5 kN swivel releases in high gear only; in low gear the release is reached at 18 N m	A. Drill in high gear only, clutch set at about 35 N m, the rule written on each trestle: R3 stays 0.42 m ³ /h; no cost. B. Clutch set at about 15 N m so it slips first in either gear: the drill cannot start the loaded train in high gear (23 N m) and hauls in low gear at about a third of the speed. C. No rule; the swivel stays the only limit, as for hand cranking	A: the swivel is the designed limit in both gears; the rule keeps the clutch as a second, earlier limit in the gear the drill is used in	A label on each trestle; the build plan already says high gear	SVD-CAL-001, D6; REVIEW round 2
5	R9 wording: option 2A meets R9 "in practice if a swap counts as maintenance", but R9 reads "72 hours of continuous operation without failure of rope, pulleys or sledges"	A. Restate R9 as "72 hours of continuous operation without failure of rope, pulleys or sledges, with ropes inspected at every shift change and a worn rope swapped as planned maintenance": R9 met in practice on paper, shown in the endurance trial. B. Keep the wording: R9 at risk until the endurance trial	A: it states what Amish chose in 2A, and the endurance trial still tests it	None	SVD-CAL-001, H; REVIEW round 2

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Pipe bore, wall and internal weld beads of the pipe used for trials	The ring is 760 mm OD with 20 mm clearance; a bead or a smaller bore changes the fit	SVD-CAL-001, A; R1
2	Winch power ratios, base bolt pattern and line rating	Handle forces assume ratios of about 13 and 40; the trestle top plates are drilled to the base	SVD-CAL-001, C; BOM line 17
3	Release load of the breakaway swivel pins bought	Sets the 2.5 kN limit that every part is sized on	SVD-CAL-001, C; R11
4	Sheave bore, bush and rating	The pin is 25 mm; the sheave must be rated at least 1,000 kg	BOM line 6
5	Friction of the jacking pads on the actual pipe surface	The ring's hold against the sheave pull assumes 0.20	SVD-CAL-001, C8
6	Rope breaking strength and splice efficiency	The factor of 6.5 assumes 18 kN and 90 % at the splice	SVD-CAL-001, C6
7	Blower curve at 525 Pa	The air figures assume 0.27 m ³ /s at the blower	SVD-CAL-001, F; R5
8	Holding force of the magnet hangers on painted or rusty pipe	They carry the duct and the cable	BOM line 22

■ Value engineering

Value-engineering target: USD 5,000 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 10,804.00 (USD 5,804.00 over the target), USD 936.50 more than before the round 2 decisions for the two portal drill drives and the spare rope set (SVD-DDR-003). Main cost drivers and savings worth trying:

- The two self-tailing winches (USD 2,300) are the largest line; reconditioned winches of the same size are often about half the new price.
- The four-gas monitors (USD 1,100), the blower and duct (USD 1,515) and the telephones (USD 860) are kit that rescue services usually already hold; borrowing them for trials removes about USD 3,500 from the prototype cost without changing the design.
- The made steel (ring, trestles, sledges) is under USD 1,500; plate rolling is the main cost there.
- Savings not worth taking: the breakaway swivels, rope and slings carry the safety case and stay at rated grades.

■ Decisions made

Pre-approval and decision quotes: Q3, Amish 2026-10-03: "i approve all of the 47 recommendations provided by you. Execute them." Q1, Amish 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." Q2, Amish 2026-10-03: "Proceed with the remaining 15 scaffolds" (under the same pre-approval).

Date	Decision	Decided by	Record
2026-10-03	Design-case pipe 800 x 10 mm, 780 mm bore	Amish, Q1 and Q2	SVD-DDR-001, item 1
2026-10-03	Ring held by jacking screws with swivel pads; nothing drilled or welded to the pipe	Amish, Q1 and Q2	SVD-DDR-001, item 2
2026-10-03	Shuttle train between a pull rope and a tail rope, in place of a circulating loop	Amish, Q1 and Q2	SVD-DDR-001, item 3
2026-10-03	Two two-speed self-tailing winches at the portal; SiltHaul capstan not used	Amish, Q1 and Q2	SVD-DDR-001, items 4, 12
2026-10-03	Breakaway swivels at 2.5 kN at both ends of the train; requirement R11 added	Amish, Q1 and Q2	SVD-DDR-001, item 5
2026-10-03	Two four-gas monitors in the kit; requirement R12 added	Amish, Q1 and Q2	SVD-DDR-001, item 6
2026-10-03	Forcing ventilation through 200 mm layflat, about 15 m ³ /min at the face	Amish, Q1 and Q2	SVD-DDR-001, item 7
2026-10-03	Sound-powered telephones with a pull-cord bell	Amish, Q1 and Q2	SVD-DDR-001, item 8
2026-10-03	Bought roll-up stretcher with a made spreader bar; casualty hauled hand over hand through winch A	Amish, Q1 and Q2	SVD-DDR-001, items 9, 10
2026-10-03	First co-design candidate to approach: India's National Disaster Response Force with the Jhansi hand miners (not agreed)	Amish, Q1 and Q2	SVD-DDR-001, item 11
2026-10-03	CalRig as the first candidate rig for the hood test and proof loads; HatchSide as the first reference for air and gas rules	Amish, Q1 and Q2	SVD-DDR-001, item 12
2026-10-03	budget_usd kept at 5,000 as a value-engineering target	Amish, Q1 ("I also accept any cost overruns or variations from the assumed scope cost.")	SVD-DDR-001, item 13
2026-10-03	Design for construction: the twelve changes of SVD-DDR-002	Amish, Q1 and Q2	SVD-DDR-002
2026-10-03	Appearance model additions for renders: pipe window and platform cropped, a 1.75 m mannequin beside the pipe	Amish, Q1 and Q2	docs/REVIEW.md, TRL 3
2026-10-03	R3 spoil haulage (open decision 1): option B, a portal drill drive on each winch socket with hand cranking as the fallback; about 0.42 m ³ /h, R3 still not met	Amish, Q3 ("i approve all of the 47 recommendations provided by you. Execute them.")	SVD-DDR-003, item 1
2026-10-03	R9 shift durability (open decision 2): option A, ropes inspected at every shift change and a spare set carried; the TRL 4 endurance trial decides whether an abrasion-resistant rope (option B) is needed	Amish, Q3	SVD-DDR-003, item 2

Date	Decision	Decided by	Record
2026-10-03	R1 600 mm stretch (open decision 3): option C, kept as a stretch until the co-design partner says whether 600 mm pipes are used for rescue	Amish, Q3	SVD-DDR-003, item 3

■ Change log

- 2026-10-03, v0.2: open decisions 1 to 3 decided as recommended (1B, 2A, 3C) and carried into the design (SVD-DDR-003); new questions 4 and 5 opened.